

Muhammad Uzair Afridi

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Professional Summary

Machine Learning and AI Engineer with 4+ years of experience turning real-world problems into production-grade AI solutions across SaaS and enterprise environments. I specialize in taking ideas from concept to deployed application: scoping the problem, designing the ML/AI approach, building the system, and operating it reliably in production. I work across machine learning, deep learning, generative AI, large language models (LLMs), and agentic AI, with hands-on expertise in RAG pipelines, NLP systems, and the end-to-end ML lifecycle.

Skills

- Python | FastAPI | Bash | JavaScript | Machine Learning | Deep Learning | Agentic AI | Generative AI | LLMs | Bash | PyTorch | Scikit-learn | TensorFlow | SQL | NoSQL | JSON | MySQL | MariaDB | Git | Hugging Face | Exploratory Data Analysis | NVIDIA GPUs |
- AWS | Gemini | CI/CD | MLOps | AIOps | MLflow | Vector Databases | Docker | LangChain | LangGraph | Pandas | NumPy | REST APIs | Data Pipelines | ETL | Model Deployment | Model Monitoring | Apache Spark | GraphQL APIs | Anomaly Detection
- Business to AI Solution Mapping | Data-Driven Decision Making | Problem Solving | Stakeholder Management | Cross-Functional Collaboration | Requirements Gathering | English, Urdu, Arabic (Basic) – professional proficiency

Experience

AI Engineer

JMM Technologies

Riyadh, KSA 06/2024 - 03/2026

- Delivered ~2.1% RMSE forecasting error on complex industrial time-series data by building and tuning an ensemble machine learning models (**Random Forest**, **XGBoost**) and **physics-informed neural networks (PINNs)** with uncertainty quantification, enabling highly data-driven decision making for plant operations.
- Achieved **85%** prediction accuracy across noisy and limited industrial datasets by owning **end-to-end ML workflows**, from data collection, preprocessing, and feature engineering to model training, evaluation, and **model deployment** in production using **Python**, **scikit-learn**, **PyTorch**, and **TensorFlow**.
- Improved enterprise knowledge retrieval accuracy by **70%** by designing and deploying **RAG** and **NLP pipelines** with **LangChain**, **LLMs**, and **vector databases (Pinecone)**, optimizing chunking strategies, embeddings, and re-ranking to power retrieval-augmented chatbots and **agentic AI assistants**.
- Enabled AI solutions to scale across **3+** enterprise workflows by implementing **MLOps** practices and robust data pipelines, containerizing models with **Docker**, exposing them via **FastAPI/REST APIs** on **AWS**, and setting up monitoring to track model performance and reliability over time.
- Accelerated stakeholder adoption of ML/AI solutions by translating business requirements into technical designs, leading stakeholder management and cross-functional collaboration with product, data, and engineering teams, and communicating results through clear dashboards, reports, and technical documentation.

Machine Learning Engineer

DevMaze

Peshawar, KPK, PAK 06/2023 - 05/2024

- Improved model performance by **25%** on regression and classification tasks by rapidly prototyping deep learning and classical ML approaches in Python with **scikit-learn** and **PyTorch**, applying feature engineering, cross-validation, and **hyperparameter tuning** to improve generalization.
- Reduced data retrieval errors by **70%** in **NLP pipelines** by designing and evaluating **RAG** systems using **LangChain** and **LLM APIs**, optimizing document chunking, indexing, and retrieval strategies on unstructured text to enhance answer accuracy for end users.
- Built fully reproducible experimentation workflows by structuring data pipelines with **Pandas** and **NumPy**, **versioning experiments**, and documenting methodology, which improved collaboration and traceability for technical and non-technical stakeholders.
- Generated actionable insights from large-scale survey and behavioral data by applying **statistical analysis**, visualization, and data-driven decision-making techniques that help stakeholders prioritize product and process improvements.
- Strengthened the ML experimentation process by contributing to internal best practices around **EDA**, feature engineering, and model evaluation metrics, and experiment logging, laying the groundwork for future **MLOps** and model governance.

Junior Python Developer

Ray Media Group

Peshawar, KPK, PAK 06/2021 - 06/2023

- Automated reporting and analytics workflows by building backend **data pipelines** and Python services that integrated with **RESTful APIs**, reducing manual reporting time and improving data freshness for business teams.

- Improved database performance and reliability by optimizing **SQL** queries on **PostgreSQL** and **MySQL**, designing efficient schemas, and cleaning legacy queries, which enhanced data-driven decision making for operations and management.
- Enhanced system stability and maintainability by performing regular code reviews, debugging production issues, and writing clear documentation, contributing to a more robust software delivery process aligned with **CI/CD** practices.
- Collaborated with cross-functional teams (content, marketing, operations) to translate business requirements into technical implementations, strengthening stakeholder management skills, and ensuring that solutions are aligned with user and business needs.
- Laid the foundation for future **ML/AI integration** by designing modular, well-structured **Python** code and APIs that could later support analytics, experimentation, and machine learning services without major architectural changes.

Education

Bachelor of Science

IMSciences University

Peshawar, KPK, PAK **09/2019 - 07/2023**

- Major in Software Engineering

Projects

- **Agentic AI Research Assistant:** Designed and deployed a fully local, privacy-focused AI research assistant using **Ollama**, **Mistral** with **Open Claw**, integrating autonomous agent workflows, real-time web search via Tavily, and automated daily insights delivery through Telegram.
- **AI CV Critic:** A smart, AI-powered tool designed to analyze and provide constructive feedback on your resume/CV. [\[LINK\]](#)
- **Smart Finance Assistant:** Build a simple yet powerful AI-powered Expense Tracker built with **FastAPI (Backend)**, **Streamlit (Frontend)**, and **Ollama (Local LLM)**. The system tracks expenses, visualizes insights through interactive charts, and uses an LLM to analyze spending patterns. [\[LINK\]](#)

Mentorship

- **GDSC & MLSA:** organizing sessions on applied ML and AI, mentoring peers, and demonstrating the ability to support the onboarding and development of new team members in technical environments.
- **Computer Science Tutor:** Programming | Data Structure and Algorithms | career advice | coding interview prep | professional portfolio

Certificates

- **Python for Machine Learning**, Dice Camp & Dice Analytics
- **Introduction to TensorFlow for AI**, Deep Learning AI
- **Supervised Machine Learning**, Deep Learning AI
- **NLP with Classification and Vector Spaces**, Deep Learning AI
- **Introduction to AI**, IBM
- **Introduction to Data Science**, IBM